

Supplementary Material

Deep Neural Networks Detect Suicide Risk from Textual Facebook Posts

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Psycho-diagnostic Tools

Suicide risk. Suicide risk was measured using the well-established Columbia Suicide Severity Rating Scale (CSSRS)¹. The CSSRS was originally developed to help clinicians structure their clinical interviews and assess the existence and severity of suicide risk with high levels of accuracy. The scale demonstrated high sensitivity and specificity scores in suicide prediction and it is considered a “diagnostic tool of choice,” both in clinical settings and in empirical research^{2,3}. Upon consultation with the principal developer of the CSSRS (Posner, personal written communication), we chose to administer the electronic self-report version of the scale, in light of the fact that the current research examined participants from a crowdsourcing platform. The electronic version of the CSSRS has been demonstrated to have psychometric validity and prediction accuracies that are comparable to the original clinician version of the scale^{4,5}.

The scale consists of six binary (yes/no) items that are presented to the participants in two parts. In the first part, participants were asked to complete Item 1 that addressed a “wish to be dead” (*“Have you wished you were dead or wished you could go to sleep and not wake up?”*) and Item 2 that

addressed "suicidal thoughts" (*"Have you actually had any thoughts of killing yourself?"*). Only if participants answered "yes" to item #2 on suicidal thoughts, they were then exposed to the second part of the scale that examined the severity of the risk. Item 3 addressed suicidal thoughts with method (*"Have you been thinking about how you might kill yourself?"*). Item 4 addressed suicidal intent (*"Have you had these thoughts and had some intention of acting on them?"*). Item 5 addressed suicide intent with specific plan (*"Have you started to work out or worked out the details of how to kill yourself? Do you intend to carry out this plan?"*), and Item 6 addressed actual suicide behaviors (*"Have you ever done anything, started to do anything, or prepared to do anything to end your life?"*). Participants who answered "yes" to this last item were then asked to indicate when they engaged in such behavior (over a year ago, between three months and a year ago, or within the last three months).

The modular structure of the CSSRS enables the extraction of two binary (yes/no) variables: a *general risk of suicide* (participants who met the criterion of the first part of the scale, that is answering "yes" to item 2) and a *high risk of suicide* (a sub-group of participants at suicide risk who also responded "yes" to at least one of the items in the second part of the scale). The total sum score of the "yes" answers to all six items serves as another indication for the severity of the suicide risk. In this study, the total score of the

CSSRS was positively correlated with all the examined risk factors (see Table 1 of the main text) and especially with depression ($r = 0.44$), thus indicating a high convergent validity of the scale.

Major Depressive Disorder (MDD). Major depression was measured using the Patient Health Questionnaire-9 (PHQ-9)⁶, a nine-item scale that targets the nine symptoms of depression described in the DSM. Each item (symptom) is scored from 0 to 3 (not at all, several days, more than half the days, and nearly every day). Given its well-established validity and high sensitivity and specificity⁷, the PHQ-9 is preferred over all other screening tools for depression⁸. The internal consistency of the scale in the current sample was high ($\alpha = .90$) and the correlation with suicide total scores was high ($r = 0.44$).

Generalized Anxiety Disorder (GAD). GAD was measured using a well-established, seven-item scale named GAD-7⁹. Each item, scored from 0 to 3 (not at all, several days, more than half the days, and nearly every day), targets one of the seven symptoms of the disorder. The total score of the scale (range 0 – 21) serves as an indication for both the existence and the severity of the disorder. The internal consistency of the scale in the current sample was high ($\alpha = .92$). The evidenced comorbidity between GAD and major depression as indicated in a bivariate Pearson, was very high ($r = 0.75$).

Depressive rumination (brooding). Depressive rumination as mentioned above is a maladaptive pattern of thinking in which people focus on their depressive feelings and enter a repetitive loop of negative thoughts^{10,11}. Specifically, the unconstructive component of this ruminative thinking, which has been shown to be strongly associated with depression was named “brooding”¹². Brooding was measured using five items rated from 1 (almost never) to 4 (almost always) from the frequently used Ruminative Responses Scale (RRS)¹³.

Respondents read a general statement about depressive events (“*People think and do many*

different things when they feel depressed”) and are asked to indicate to what extent they engage in a given response. An example for a brooding response is: “*Think about a recent situation, wishing it had gone better.*” The internal consistency of the 5 brooding items, in the current sample was good ($\alpha = .82$) and the correlation with depression was high ($r = 0.61$).

Excessive worrying. A second pattern of negative thinking is excessive and subjectively uncontrollable worries about the future¹⁴. To assess excessive worrying patterns, we used the Penn State Worry Questionnaire (PSWQ)¹⁵. The PSWQ is a well-established research tool¹⁶ that comprises 16 items, rated on a five-point scale (1 = not as all typical of me, 5 = very typical of me). The items address various aspects of pathological worry including its excessiveness (e.g., “*Many situations make me worry*”) and the subjective feeling of uncontrollability (e.g., “*Once I start worrying, I cannot stop*”). The internal consistency of the PSWQ in the current sample was high ($\alpha = .96$) and the correlation with depression was high ($r = 0.55$).

Loneliness. Experiences of loneliness were measured using the 10-item version of the UCLA-Loneliness Scale¹⁷. The items, rated from 1 (Never) to 4 (Always), encompass various aspects of loneliness experiences (e.g., “*How often do you feel that you lack companionship*”). This version of the scale demonstrated high levels of convergent validity and internal consistency (Elphinstone, 2018). The internal consistency of the scale in the current sample was high ($\alpha = .92$) and the correlation with depression was high ($r = 0.57$).

Low satisfaction with life. The general sense of satisfaction with life was measured using the Satisfaction With Life Scale (SWLS)¹⁸. This short scale comprises five items, rated from 1 (strongly disagree) to 7 (strongly agree). All items are formulated in a positive manner (e.g., “*The conditions of my life are excellent*”). Although we were interested in low satisfaction with life, we kept the original positive style of the scale to “break” the overall negative atmosphere of the research and to promote participants’ attentiveness along the

research. The SWLS demonstrated good psychometric characteristics¹⁹ and moderate-strong negative relationships with depression^{20,21}. The internal consistency of the scale in the current sample was high ($\alpha = .93$) and the negative correlation of this positive scale with depression was high ($r = -0.53$).

Personality traits. Personality traits were assessed using the short version of the Big Five Inventory (BFI)²². The BFI-10 includes ten items that target the five clusters of personality traits, originally formulated in the standard 44-item BFI: Extraversion, Neuroticism, Openness to Experience, Agreeableness and Conscientiousness²³. Each trait in the BFI-10 is measured by only two items that are rated from 1 (disagree strongly) to 5 (agree strongly). The BFI-10 achieved high levels of reliability and validity²² and is currently widely used in research settings. Consistent with the literature on depression, the correlation between the personality trait of neuroticism and depression was high ($r = 0.51$).

The convergent validity of the psychosocial scales was high. As expected, the total score of the suicide scale was positively correlated with all the risk factors examined in the study and especially with depression ($r = 0.49$). Consistent with the literature on depression described above, the comorbidity between depression and anxiety was very high ($r = 0.75$) and the four psychosocial risk factors (i.e., brooding, excessive worry, loneliness, and low satisfaction with life) were strongly correlated with depression (Pearson's r ranging from 0.53 to 0.61). The personality trait of neuroticism was also strongly correlated with depression ($r = 0.49$).

Ethical Considerations

Crowdsourcing-based suicide research involves an ethical challenge: how to safeguard the well-being of suicidal participants, without the possibility of face-to-face interactions? To address this ethical challenge, the first author (a clinical psychologist) cooperated with a consortium of experts to develop an online suicide research protocol. Prior to consenting to

participate in the study, participants were informed that if their responses would indicate some form of suicidal risk, we would contact them through the data collection platform. Each participant who met the CSSRS criterion for general suicide risk (i.e., suicidal thoughts with or without a specific method or a concrete plan) then received a designated letter in which we encouraged them to seek help and provided them with a list of available “hotlines” and national mental health services. The complete description of the protocol and the ethical considerations made in the current research are available by the first author upon request.

Data Quality

In light of recent concerns regarding the quality of crowdsourcing-based data, we applied a newly developed rigid data quality assurance protocol²⁴. The inclusion criteria were: having a Facebook account and having previous proven experience in MTurk-based studies. Proven experience was defined as past completions of at least 100 MTurk tasks, with a minimum of 95% success rate. To avoid bogus responses, we limited the participation to US residents and excluded users with suspicious Internet Protocol (IP) addresses. To ensure the quality of the unsupervised responses, we implemented a designated inattentiveness scale that comprised eight hidden attention checks. These checks included four types of data-quality measurements (i.e., “infrequency items,” “time measurements,” “person-total correlation,” and “long string analysis”), which were embedded in the various self-report scales of the study²⁴.

Loss functions of the ANN-based Models

As illustrated in Figure 1 of the main article, the

Single-Task Model (STM) consisted of an input and an output layer, which are connected by a set of fully-connected layers. In contrast, the Multi-Task Model (MTM) contained three additional hierarchically organized auxiliary layers: Facebook content $\rightarrow$ personality traits $\rightarrow$ psychosocial risks $\rightarrow$ psychiatric disorders $\rightarrow$ suicide risk.

The loss function of the STM models is the *binary cross-entropy*:

$$1. L_{suicide} = -\frac{1}{N} \sum_{i=1}^N y_i \log \log(p(y_i)) + (1 - y_i) \log \log(1 - p(y_i))$$

Where N is the number of training examples, y_i indicates whether participant i belongs to the suicide group ($y_i = 1$) or not ($y_i = 0$) according to the ground truth, and $p(y_i)$ indicates the probability of y_i as predicted by the model.

The loss function of the MTM is the sum of the output layer's and the auxiliary layers' loss functions:

$$2. L = L_{suicide} + L_{aux}$$

Where $L_{suicide}$ is the binary cross-entropy loss function like before, and L_{aux} is the sum of all *mean squared errors* (MSEs) calculated for each of the auxiliary variables in the set $A = \{\text{Depression, Anxiety, Brooding, Worry, SWL, Lonely, Open, Conscientious, Extravert, Agreeable, Neurotic}\}$:

$$3. L_{aux} = \sum_{a \in A} -\frac{1}{2N} \sum_{i=1}^N \left(y_i^{(a)} - \hat{y}_i^{(a)} \right)^2$$

where N is the number of training examples, $y_i^{(a)}$ is a continuous variable representing the ground truth score of the

auxiliary-variable a for subject i , and $\hat{y}_i^{(a)}$ is the predicted score for this variable according to the model.

ANN-based Models – Parameter Estimation (Learning)

The optimization of the model was conducted with batch sub-gradient descent (batch-size of 32), using the back-propagation algorithm²⁷ and the RMSProp optimizer²⁸ with a momentum parameter of 0.9. The hyper-parameters of the models were tuned using a grid-search method on the development data, and the model with the highest development data AUC score for suicide risk (general or high) was selected. These hyper-parameters included the number of fully connected layers $\{1, 2, 3\}$ (note that all the sub-networks of the MTM had the same number of fully connected layers), the number of neurons in each layer $\{16, 32, 64, 128, 256, 512, 1024\}$, and the type of the activation function $\{\textit{hyperbolic tangent}, \textit{sigmoid}\}$. The hyper-parameters of the optimization algorithm were the learning rate $\{0.001, 0.005, 0.01, 0.05\}$, and the number of epochs $\{1000, 2500, 5000\}$.

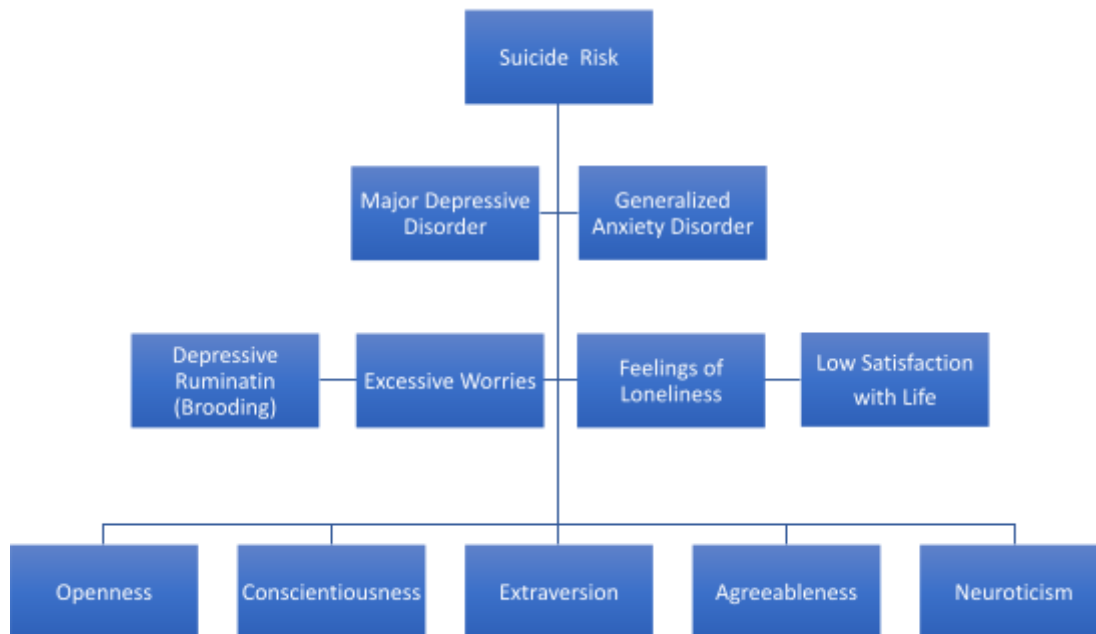
Following, we describe the selected hyper-parameters of the STM and the MTM of the first fold of the 5-folds, cross validation process. All other hyper-parameters are available upon request from the authors. The selected hyper-parameters of the General risk STM included: 3 fully connected layers, 32 neurons, an activation function of *hyperbolic tangent*, a learning rate of 0.01 and 2,500 epochs. The selected hyper-parameters of the high risk STM included the same parameters but with 5,000 epochs.

The selected hyper-parameters of the general risk MTM included: 2 fully connected layers, 16 neurons, an activation function of *Sigmoid hyperbolic tangent*, a learning rate of 0.001 and 5,000 epochs. The selected hyper-parameters of the high risk MTM: 3 fully connected layers, 16 neurons, an activation Sigmoid function, a learning rate of 0.001 and 1,000 epochs.

Four Possible Classes of Suicide Risk Predictions

The continuous general risk score that each participant received from the MTM (the best performing model) was transformed into a binary general suicide risk label (positive/negative). The threshold from the ROC curve that was chosen for this transformation was the one that returned the maximum ratio between the True Positive and the False Positive rates. Based on this threshold, users were classified into four groups: True Positive, in which a suicidal user is correctly detected (true) by the model as suicidal (positive); False Positive, in which a non-suicidal user is incorrectly detected (false) as suicidal (positive); True Negative in which a non-suicidal user is correctly determined (true) as not suicidal (negative); and False Negative in which a suicidal user is incorrectly determined by the model (false) as non-suicidal (negative).

Figure A. Illustration of the hierarchical "pyramid" of risk factors for suicide



Note: The bottom of the proposed pyramid consists of the big five personality traits (i.e., openness; conscientious; extraversion; agreeableness; and neuroticism). The middle layers consist of the psychosocial risk factors (i.e., depressive rumination, worries, loneliness, and low satisfaction with life) and the psychiatric disorders (i.e., depression and anxiety), and the top layer consists of the predicted output, which is the two types of binary suicide variables (i.e., general and high suicide risk).

Suicide detection using BERT

Table A provides a comparison of the results of the STM and MTM models, between the case where the text representation is made by ELMo (the text representation method that was employed in the main study) and the case where the text is represented by the recent attention-based BERT model (Bidirectional Encoder Representations from Transformers)²⁹. The similarities between the two cases included an equivalent range of AUC scores and improved predictions of the theory-driven MTM. On average, the MTM produced higher AUC scores than the STM, both in the general risk case (Mean difference = .073, 95% CI: .030, .139) and the high risk case (Mean difference = .076, 95% CI: .053, .100). The difference between the two cases included better BERT performance on the high risk group compared to the general risk group (an opposite phenomenon is observed with ELMo). The overall similar patterns indicate that the main conclusions of the research, and particularly the one about the importance of theory-driven multi-task modeling for suicide risk prediction, are independent of the specific text representation method employed by the model.

Table A. Detection performance of the models using BERT Task

	General suicide risk		High suicide risk	
Model	STM	MTM	STM	MTM

Average AUC scores using ELMo	.621 95% CI: .576, .657	.746 95% CI: .727, .765	.629 95% CI: .606, .660	.697 95% CI: .690, .707
Average AUC scores using BERT	.599 95% CI: .569, .660	.672 95% CI: .608, .708	.679 95% CI: .638, .727	.744 95% CI: .676, .780

Note: STM = Single Task Model; MTM = Multiple Tasks Model; AUC = Area Under the receiver operating characteristic Curve; Average AUC scores = The average scores of the five AUC scores that were obtained in the cross-validation analyses.

Table B. Term Frequency Inverse Document Frequency (TF-IDF)

The following table presents the 100 most characteristic words that best distinguished each one of the four classes from the others in the general risk MTM of the fold (from the 5-folds cross validation process) that resulted in the best AUC score.

	True Positive	False Positive	True Negative	False Negative
1	anymore	president	lord	fight
2	lose	husband	shall	ufb
3	kinda	season	blessed	office
4	probably	games	king	daddy
5	mad	gun	gift	fire
6	poor	strong	james	system
7	positive	wonder	christ	email
8	room	cat	church	experience
9	pretty	continue	missed	nobody
10	sitting	update	drink	shows
11	top	news	loving	ufd
12	cheese	damn	anybody	just
13	air	paid	god	america
14	eating	writing	monday	war
15	sick	none	wedding	none
16	okay	words	faith	questions
17	pain	talking	christmas	king
18	cry	change	jesus	ya
19	actually	ass	coffee	etc
20	instead	hot	prayer	pray
21	seriously	cold	however	like
22	easy	test	prayers	day
23	bit	watched	version	bus
24	reason	within	history	prayers
25	game	children	kill	one
26	clothes	fb	wishes	can
27	mother	asking	answer	happy
28	daughter	knew	state	know
29	hurt	grow	comment	get
30	worst	cancer	law	whats
31	bad	blood	important	will
32	account	shit	heaven	time
33	felt	red	mental	five
34	theyre	yesterday	father	texas
35	wow	dad	dr	sister
36	enjoy	holiday	ready	soul
37	lady	country	son	running

38	cut	near	pass	street
39	sleep	look	spirit	coming
40	fuck	loved	ufc	green
41	supposed	weeks	email	yeah
42	bed	looked	peace	american
43	pizza	funny	line	fine
44	quite	text	thanksgiving	along
45	gets	around	perfect	attention
46	guess	support	ufb	close
47	drive	relationship	war	human
48	door	course	save	died
49	thinking	putting	lol	company
50	surgery	wanted	together	problems
51	gonna	seems	dear	aint
52	literally	found	thank	business
53	thats	couple	just	prayer
54	sleeping	world	great	share
55	bitch	daily	sunday	area
56	cream	john	happy	listen
57	heart	looks	working	bit
58	wonderful	ask	fall	water
59	big	several	day	truly
60	arent	posts	cause	missing
61	might	moving	st	go
62	fucking	half	given	character
63	hospital	age	today	page
64	told	seeing	help	people
65	sad	company	men	store
66	doesnt	longer	love	love
67	couldnt	kept	choose	local
68	wall	months	friends	group
69	favorite	tv	brother	learned
70	taking	florida	others	retweeted
71	cleaning	high	holy	send
72	stupid	others	time	busy
73	nap	away	like	good
74	ugh	given	city	gonna
75	start	kids	everyone	song
76	entire	women	giving	now
77	brain	place	roll	anybody
78	car	hand	please	im
79	wear	sit	florida	park
80	times	weekend	mother	sorry
81	dinner	hey	copy	see
82	play	fear	question	pass
83	story	run	wife	gas
84	Isnt	voice	know	birthday
85	calling	dark	friend	wishes

86	white	called	forget	asking
87	spent	spend	cleaning	end
88	mind	eyes	teacher	books
89	online	learn	get	lives
90	hopefully	light	group	miss
91	ice	history	pray	everyone
92	making	true	child	jesus
93	rest	saw	can	missed
94	feet	body	go	dark
95	order	heard	known	figure
96	cute	open	safe	wedding
97	understand	happened	one	entire
98	type	move	busy	today
99	sure	write	american	came
100	summer	due	lets	forget

Note: This table presents the 100 most characteristic words that best distinguished each one of the four classes (True Positive, True Negative, False Positive, and False Negative), from the others, using *Term Frequency Inverse Document Frequency (TF-IDF)*.

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